

5	(a) (i) force proportional to the product of the two/point charges and inversely proportional to the square of their separation	B1	
		B1	[2]
	(ii) 1. force radially away from sphere/to right/to east	B1	[1]
	2. (maximum) at/on surface of sphere <i>or</i> $x = r$	B1	[1]
	3. $F \propto 1/x^2$ <i>or</i> $F = q_1q_2/(4\pi\epsilon_0x^2)$	C1	
	ratio = 16	A1	[2]
	(b) $E = q/(4\pi\epsilon_0x^2)$ <i>or</i> $E \propto q$	C1	
	maximum charge = $(2.0/1.5) \times 6.0 \times 10^{-7}$ = $8.0 \times 10^{-7} \text{ C}$	C1	
	additional charge = $2.0 \times 10^{-7} \text{ C}$	A1	[3]